

 Section: Our Dynamic Universe

 Topic: The Expanding Universe

(a) Comment on the lyrics using your knowledge of physics

✔ 3 marks

Lyrics:

“A fraction of a second and the elements were made...
It’s expanding ever outward but one day it will cause the stars to go the other way...”

✔ **Commentary based on the physics content:**

1. “A fraction of a second and the elements were made...”

- This refers to the process of **nucleosynthesis** in the early universe.
- In the first few minutes after the Big Bang, **protons and neutrons combined** to form light nuclei like hydrogen, helium and small amounts of lithium.
- This event is known as **primordial nucleosynthesis**, which occurred during the first few minutes, not strictly a *fraction of a second*, but close enough for poetic license.

 Mark for: recognition of **nucleosynthesis** and the formation of **light elements** in early universe.

2. “It’s expanding ever outward...”

- This is consistent with current observations that the universe is **expanding**, supported by **redshift** in the light from distant galaxies and described by **Hubble’s Law**.

 Mark for: recognising **Hubble expansion**.

3. “...but one day it will cause the stars to go the other way...”

- This line suggests a possible **future contraction** of the universe (i.e. the **Big Crunch** scenario).
- However, current data (e.g. from Type Ia supernovae) shows that the expansion of the universe is **accelerating**, due to **dark energy**, so the idea of contraction is now considered **unlikely**.

 Mark for: recognising the **speculative nature** of universal contraction and contrast with current understanding.

✔ **Final 3-mark answer example:**

The line about elements being formed refers to nucleosynthesis in the early universe, when light elements like hydrogen and helium were created. The reference to expansion reflects the observed expansion of the universe, as seen in redshifted light from galaxies. The suggestion that the stars will “go the other way” refers to a possible future collapse of the universe, but current evidence suggests expansion is accelerating, making a collapse unlikely.

 Revision Tips

- Light element formation = **primordial nucleosynthesis**
- **Redshift** and **Hubble’s Law** support expansion
- **Big Crunch** is a theoretical scenario not supported by current observations
- Expansion is currently **accelerating** due to **dark energy**